BJT DC bias II

Cheat sheet · Day 08 · Sat 31 Oct · pack E5

Reference-sheet lines

COPY THIS ONTO YOUR PRINTED REFERENCE SHEET

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BJT DC BIAS II (E5)
STATE LOOP: guess -> analyse -> check -> redo
 active: |VBE|=0.7, IC=bIB, IE=(b+1)IB    check IB>0, |VCE|>0.2
 sat:    |VBE|=0.7, |VCE|=0.2, IE=IB+IC   check IB,IC>0, IC<bIB (quote IC/IB)
 cutoff: IB=IC=IE=0                        check |VBE|<0.7
 VCE<=0.2 (negative!) -> sat ;  IB<=0 -> cutoff
 grounded emitter: IB same in both attempts, IC,sat=(VCC-0.2)/RC
 RE present: base & collector loops coupled in sat -> solve 2x2

GENERAL:  IB = (VTh - VBE) / (RTh + (b+1)RE)
 beta range: AVOID sat -> use b_max ; GUARANTEE sat -> use b_min (edge: IC=bIB, VCE=0
 negative R from inequality -> "no value works", explain

COLLECTOR FEEDBACK: IB=(VCC-VBE)/(RF+(b+1)(RC+RE)); RC carries IE
   VCE=VCC-IE(RC+RE)   (slides use b(RC+RE): IC~IE)
EMITTER FOLLOWER: IB=(VEE-VBE)/(RB+(b+1)RE); VCE=VEE-IE*RE
COMMON BASE: IE=(VEE-VBE)/RE; VCB=VCC-IC*RC; VCE=VEE+VCC-IE(RC+RE)

PNP: VB=VE-0.7; IB, IC OUT; IE IN; check VEC>0.2. Draw arrows first.
TWO STAGES: solve stage with own base loop first; shared node;
 npn 2nd stage: RC1 carries IC1+IB2 ; pnp 2nd stage: RC1 carries IC1-IB2
 diode-connected (B=C): always active, drops 0.7 V, its beta irrelevant
APPROX DIVIDER valid only if b*RE >= 10 R2 (never with RE=0)

DESIGN (VDB): VE=0.1VCC; RE=VE/IC; RC=(VCC-VCE-VE)/IC;
 R2 <= b_min RE/10; VB=VE+0.7; R1=R2(VCC-VB)/VB; standard values -> re-check exact
EARLY: iC=IS e^(vBE/VT)(1+vCE/VA); ro=(VA+VCE)/IC ~ VA/IC
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